Tab3a W Hdiyi

By

Georgio Mounif Chawiche

CSCI-490 – Information System Development

Abstract

This project introduces a personalized gift e-commerce website that allows users to customize items like mugs, rocks, and t-shirts with their own images or text. Built with HTML, CSS, JavaScript, Node.js, and MySQL, the platform features real-time previews using HTML5 Canvas, secure checkout, and mobile-friendly design. It includes user accounts, order tracking, and an admin dashboard for managing inventory and sales. The goal is to deliver a smooth, intuitive shopping experience that blends creativity with functionality.

Project Overview

The Website allows customers to:

- Browse gift items with detailed descriptions and images
- Customize products with text and images
- Preview designs before purchase
- View store location and product reviews
- Enjoy a responsive, user-friendly experience
- View and receive sales.

Motivation

Current online stores lack:

- Personalized printing options
- Real-time customization previews
- Local access to custom gift services
- Smooth mobile experiences

Key Features

- Product catalog with customizable items (Browse through products)
- HTML5 Canvas-based customization tool (In order to Customize specific gifts)
- Shopping cart and secure checkout
- User account system with order tracking
- Admin dashboard for inventory and sales management

Technology Stack

- Software: Visual Studio Code, XAMPP, MySQL
- Frontend: HTML, CSS, JavaScript
- Backend: Node.js
- Database: MySQL, PHP
- Customization: HTML 5 canvas for dynamic reviews

Challenges and solutions

Challenge	Solutions
Mobile Responsiveness	**Mobile-first design with touch-friendly controls**
Image Upload Quality	**Validation, resolution checks, auto-compression**
Real-Time Preview	**HTML5 Canvas overlay for dynamic product visualization**

Expected Outcomes

- Fully functional e-commerce platform
- Interactive customization experience
- Secure user and admin systems
- Mobile-friendly design
- Admin Sales Dashboard A backend interface where admins can create sales campaigns, apply discounts, and monitor profit margins in real time
- Customers can leave feedback on products, helping others make informed decisions and improving product quality over time.
- Admins can update stock levels, receive low-stock alerts, and manage product availability dynamically.

UML Use Case Diagram

To visualize the functional requirements of the system, multiple use case diagrams is used. Each diagram represents a different application domain and highlights the interactions between users (actors) and system functionalities.

Main Actors

1. Customer

a) Registered Costumer

An user who is registered and using an account and have accessibility for such things

b) New Customer (guest)

A guest whom entered the website without signing up

2. Admin

An admin has the accessibility for the products and manage sales

3. System

It manages to keep the website working properly

Use case by Actors

New Customer Use Cases

- Go to the Registration page and create new account
- Can browse through products

Registered Customer Use Cases

- Go to the Registration page and log in with his account
- Browse through the website and view products
- Extends: Read Reviews of the products
- Extends: After reviewing the product if the customer wants he/she can add the specific product to cart
- Customize specific products
    - Extends: Adds a specific text on the product that the user wrote
    - Extends: Adds a specific photo on the product that the user uploaded
    - Includes: View the final result of the product
    - Extends: add the customized to the cart

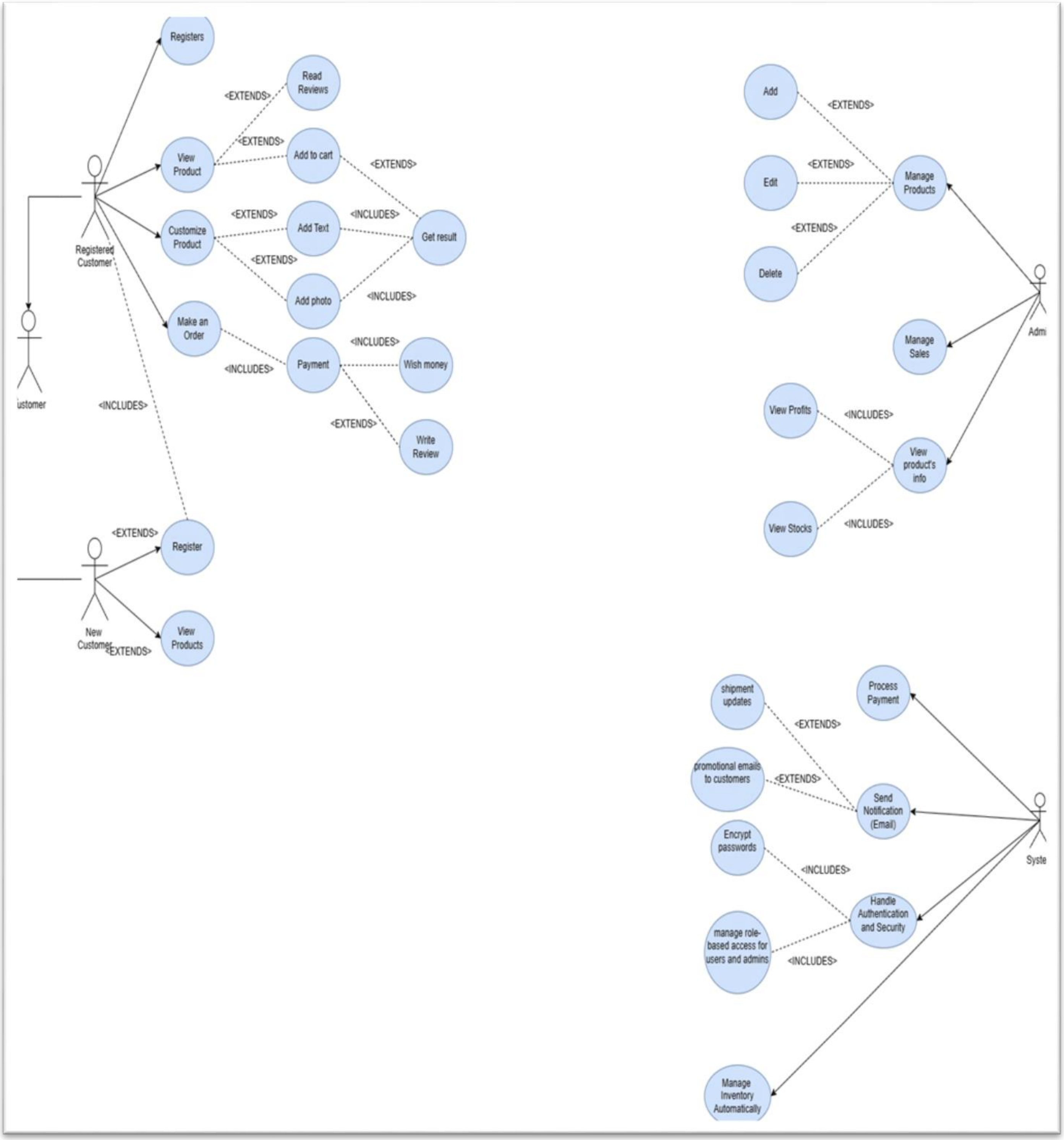
- Make an order after confirming the products the customer wants
    - Includes: Pay for the products by Wish Money
    - Extends: Leave a review of the product the customer bought after the purchase is completed

Admin Use Cases

- Manages Products
    - Extends: Adds a product to the website
    - Extends: Edit some specific information about the products
    - Extends: Delete specific products from the website
- Manage Sales by selecting a specific product and put it on sale at a specific value
- View product's information
    - Extends: View Profits of the sold products
    - Extends: View the remaining stocks available and also gets notified if the stock level of an specific product is low

System Use Cases

- Process Payment
- Send Notification via Email
    - Extends: Sends about shipment updates of an order
    - Extends: Sends promotional emails to customers when a sale is happening
- Handle authentication and security
    - Includes: Encrypt passwords
    - Includes: Manage role-based access for users and admins
- Manage inventory Automatically


Registers
Read Reviews
<EXTENDS>
<EXTENDS>
View Product
Add to cart
<EXTENDS>
Registered Customer
Customize Product
<EXTENDS>
Add Text
<INCLUDES>
Get result
<EXTENDS>
Add photo
<INCLUDES>
Make an Order
<INCLUDES>
Payment
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Wish money
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Write Review
ustomer
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Register
New Customer
<EXTENDS>
View Products
Add
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Edit
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Manage Products
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Manage Sales
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View product's info
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shipment updates
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promotional emails to customers
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Send Notification (Email)
Encrypt passwords
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Syste
Handle Authentication and Security
manage role-based access for users and admins
<INCLUDES>
Manage Inventory Automatically

Use Case Narrative

In this part we will explain about all mentioned use case of the actors with a short description

1.View Products

Use-Case Name:	View Product List
Use-Case ID:	EC-VPL.001.00
Use-Case Type:	Business Requirement
Priority:	low
Source	Requirement – EC-FR001
Primary Business Actor:	Customer (New or Registered)
Other Participating Actors:	Customer (New or Registered)
Description:	Allows customers to browse available products. • New customers see basic info. • Registered customers can read reviews, add to cart, and customize products.
Precondition:	Products must be available and visible in the system catalog.

Trigger:	Actor accesses “Products” section via homepage, search, or category link.	
Typical Course of Events:	Actor Action	System Response
	Step 1: Actor selects “View Products” from homepage or menu Step 3: Actor browses product listings Step 5: Registered actor clicks “Read Reviews” or “Add to Cart” Step 7: Actor selects “Customize Product” Step 9: Actor previews result	Step 2: System verifies actor type (New or Registered) Step 4: System displays product cards with name, image, price, and options Step 6: System loads reviews or adds item to cart Step 8: System opens customization panel with text/photo options Step 10: System generates final preview and confirms customization
Conclusion:	The use case ends when the actor views or customizes a product. Registered users may proceed to add items to cart, place orders, or write reviews. Admins and system processes may track engagement and inventory in the background.	

2.Registers

Use-Case Name:	Register
Use-Case ID:	EC-REG.002.00
Use-Case Type:	Business Requirement

Priority:	High	
Source	Requirement – EC-FR002	
Primary Business Actor:	New Customer	
Other Participating Actors:	Registered Customer, System	
Description:	Enables new users to create an account and existing users to update credentials or re-authenticate. • New customers provide personal info and credentials. • Registered users may re-register for role updates or password recovery.	
Precondition:	Actor must not be currently authenticated (for new registration).	
Trigger:	Actor clicks “Register” from homepage, login page, or contextual prompt.	
Typical Course of Events:	Actor Action	System Response
	Step 1: Actor selects “Register” option Step 3: Actor enters name, email, password, and optional preferences Step 5: Actor submits form	Step 2: System displays registration form Step 4: System validates input and checks for existing account Step 6: System creates account and stores encrypted credentials

	Step 7: Actor receives confirmation message or email	Step 8: System redirects to login or dashboard based on role
Conclusion:	The use case ends when the actor successfully registers and gains access to the system. Registered users may proceed to view products, customize items, or place orders. Admins may later verify or manage user accounts.	

3.Customize Product

Use-Case Name:	Customize Product
Use-Case ID:	EC-CUS.003.00
Use-Case Type:	Business Requirement
Priority:	High
Source	Requirement – EC-FR003
Primary Business Actor:	Registered Customer
Other Participating Actors:	System
Description:	Allows registered users to personalize products before purchase. • Users can add custom text or upload images. • A preview is generated to show the final result

Precondition:	Actor must be logged in and viewing a customizable product.	
Trigger:	Actor clicks “Customize” on a product detail page or card.	
Typical Course of Events:	Actor Action	System Response
	Step 1: Actor selects “Customize Product”	Step 2: System opens customization interface
	Step 3: Actor chooses “Add Text” option	Step 4: System displays text input field and font/style options
	Step 5: Actor chooses “Add Photo” option	Step 6: System prompts for image upload and placement
	Step 7: Actor clicks “Get Result”	Step 8: System generates and displays preview of customized product
	Step 9: Actor confirms customization	Step 10: System saves customization and prepares product for checkout
Conclusion:	The use case ends when the actor finalizes the customization and either adds the product to cart or proceeds to order. The system stores the design and ensures it’s linked to the order. Admins may later review customizations for quality or fulfillment	

4.Make An Order:

Use-Case Name:	Make an Order	
Use-Case ID:	EC-ORD.004.00	
Use-Case Type:	Business Requirement	
Priority:	High	
Source	Requirement – EC-FR004	
Primary Business Actor:	Registered Customer	
Other Participating Actors:	System	
Description:	Enables registered users to place orders for products in their cart. • Includes payment processing and optional use of store credit or gift cards (“Wish Money”).	
Precondition:	Actor must be logged in and have items in the cart.	
Trigger:	Actor clicks “Checkout” or “Place Order” from cart or product page.	
Typical Course of Events:	Actor Action	System Response
	Step 1: Actor reviews cart and clicks “Place Order” Step 3: Actor selects payment method and enters details	Step 2: System verifies cart contents and user authentication Step 4: System processes payment securely

	Step 5: Actor applies "Wish Money" or gift card if available Step 7: Actor confirms order Step 9: System sends notification (email	Step 6: System deducts applicable credit and updates total Step 8: System generates order ID and sends confirmation Step 10: System updates inventory and triggers shipment workflow
Conclusion:	The use case concludes when the order is successfully placed and confirmed. The system updates inventory, sends notifications, and prepares for delivery. Admins may later track sales, view profits, or manage fulfillment	

5.Manage Products:

Use-Case Name:	Manage Products
Use-Case ID:	EC-MPR.005.00
Use-Case Type:	Business Requirement
Priority:	High
Source	Requirement – EC-FR005

Primary Business Actor:	Admin	
Other Participating Actors:	System	
Description:	Allows Admin to oversee the product catalog. • Includes adding new products, editing existing ones, and removing outdated or unavailable items.	
Precondition:	Admin must be authenticated with appropriate access rights.	
Trigger:	Admin accesses “Product Management” section from dashboard or admin panel.	
Typical Course of Events:	Actor Action	System Response
	Step 1: Admin selects “Manage Products” Step 3: Admin clicks “Add Product” Step 5: Admin clicks “Edit” on a product Step 7: Admin clicks “Delete” on a product Step 9: Admin saves changes	Step 2: System displays product list with action buttons Step 4: System opens form for product details Step 6: System loads editable fields for selected product Step 8: System prompts for confirmation and removes product Step 10: System updates catalog and syncs with inventory

Conclusion:	The use case ends when the Admin successfully adds, edits, or deletes products. The system updates the catalog and reflects changes across customer-facing pages and inventory tracking.

6. View Product's Info:

Use-Case Name:	View Product's Info
Use-Case ID:	EC-VPI.007.00
Use-Case Type:	Business Requirement
Priority:	Medium
Source	Requirement – EC-FR007
Primary Business Actor:	Admin
Other Participating Actors:	System
Description:	Allows Admin to access detailed product data. • Includes name, price, stock level, sales history, and performance metrics. • Supports decision-making for inventory and marketing.
Precondition:	Admin must be authenticated and have access to product management or analytics.
Trigger:	Admin clicks on a product from the dashboard, stock list, or profit report

Typical Course of Events:	Actor Action	System Response
	Step 1: Admin selects a product from stock or sales view Step 3: Admin views product details Step 5: Admin filters or compares with other products	Step 2: System verifies access rights and loads product data Step 4: System displays info including name, price, quantity, and performance Step 6: System updates view with selected filters or comparison metrics
Conclusion:	The use case ends when the Admin reviews product information for decision-making. This data may inform restocking, promotions, or product retirement. It also supports deeper insights in View Profits and View Stocks workflows	

System Design

- ER-Diagram

This section presents the Entity-Relationship Diagram (ERD) for the Tab3a w Hdiyi platform. The diagram illustrates the core entities, their attributes, and the relationships between them. It serves as the foundation for database design and backend logic.

-Entities and Attributes:

1. Users

Represents all registered users of the system, including customers and admins.

Attributes	Data Types	Description
USER_ID	INT(11),AUTO_INCREMENT,PK	Unique identifier for each user
NAME	VARCHAR(25)	Full name of the user
EMAIL	VARCHAR(50),UNIQUE	User’s email address
PASSWORD	VARCHAR(255)	Encrypted password

TYPE	ENUM ('customer','admin','delivery')	Role of the user

2.ORDERS

Stores all order transactions placed by users.

Attribute	**Data Types**	**Description**
ORDER_ID	INT(11),AUTO_INCREMENT,PK	Unique order identifier
TOTAL_AMOUNT	DECIMAL(10,2)	Final amount including delivery fee
STATUS	VARCHAR(50)	Current status of the order
CREATED_AT	DATETIME	Timestamp of order creation
ORDER_DATE	DATE	Date of the order
LOCATION	VARCHAR(100)	Delivery location
PHONE	VARCHAR(20)	Contact number
USER_ID	INT(11),FK =>USERS	Links order to user

3.Orders

Contains all items available for purchase.

Attributes	Data Type	Description
PRODUCT_ID	INT(11),AUTO_INCREMENT,PK	Unique product identifier
PRODUCT_NAME	VARCHAR(100)	Name of the product
DESCRIPTION	TEXT	Detailed description
IMAGE	LONGBLOB	Binary image data
BUY_PRICE	DECIMAL(10,2)	Purchase cost for admin
SELL_PRICE	DECIMAL(10,2)	Selling price for customers
CATEGORY	VARCHAR(50)	Product category
STOCK_QUANTITY	INT(11)	Available stock
IS_CUSTOMIZABLE	TINYINIT(1),DEFAULT(0)	Flag for customization availability

4.CONTAINS

Represents the items included in each order.

Attributes	Data Types	Description
ITEM_ID	INT(11),AUTO_INCREMENT,PK	Unique item identifier
ORDER_ID	INT(11),FK=>ORDERS	Links item to order
PRODUCT_ID	INT(11),FK=>PRODUCTS	Links item to product
QUANTITY	INT(11)	Quantity ordered
UNIT_PRICE	DECIMAL(10,2)	Price per unit at time of order

5.COSTUMIZATIONS

Tracks customization actions performed by users.

Attributes	Data Types	Description
CUSTOMIZATION_ID	INT(11),AUTO_INCREMENT,PK	Unique customization ID
TYPE	VARCHAR(50)	Type of customization (text, image, etc.)
USER_ID	INT(11),FK=>USERS	User who created customization
PRODUCT_ID	INT(11),FK=>PRODUCTS	Product being customized

6.REVIEW

Stores customer feedback on products.

Attributes	Data Types	Description
USER_ID	INT(11),FK=>USERS	User who wrote the review
PRODUCT_ID	INT(11),FK=>PRODUCTS	Product being reviewed
REVIEW_TEXT	TEXT	Written feedback
RATING	INT(11)	Numerical rating (e.g., 1–5 stars)
REVIEW_DATE	DATE	Date of review submission

Relational Diagram Explanation:

The relational diagram illustrates how the entities in the e-commerce system are connected through primary keys (PK) and foreign keys (FK). It ensures data integrity and supports the business logic of orders, products, reviews, and customizations.

1.USERS<=>ORDERS

- **Relationship:** One-to-Many
- **PK/FK Link:** USERS.USER_ID → ORDERS.USER_ID
- **Meaning:** A single user can place multiple orders, but each order belongs to exactly one user.

2.ORDERS<=>CONTAINS<=>PRODUCTS

- **Relationship:** Many-to-Many (resolved via CONTAINS table)
- **PK/FK Links:**
    - `ORDERS.ORDER_ID` → `CONTAINS.ORDER_ID`
    - `PRODUCTS.PRODUCT_ID` → `CONTAINS.PRODUCT_ID`

- **Meaning:** Each order can contain multiple products, and each product can appear in multiple orders. The CONTAINS table captures this relationship with additional attributes like `QUANTITY` and `UNIT_PRICE`.

3.USERS<=>REVIEWS<=>PRODUCTS

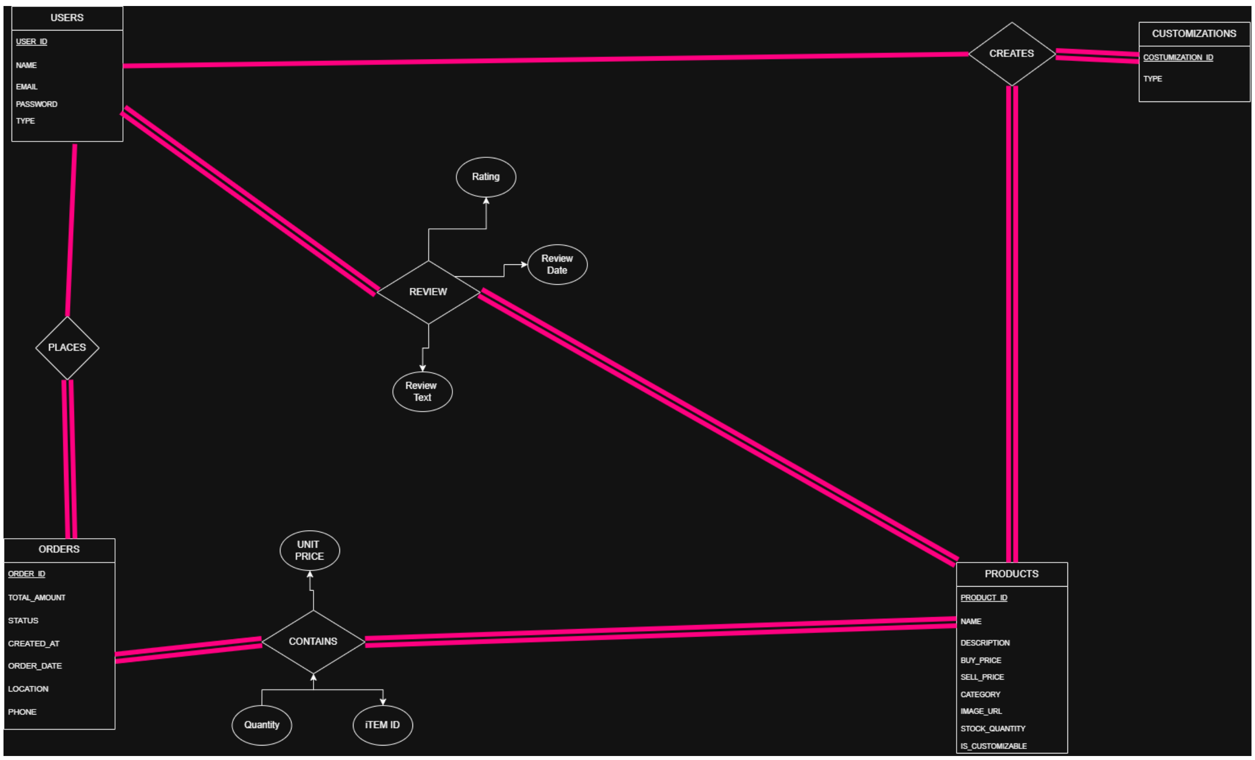
- **Relationship:** Many-to-Many (resolved via REVIEW table)
- **PK/FK Links:**
  - `USERS.USER_ID` → `REVIEW.USER_ID`
  - `PRODUCTS.PRODUCT_ID` → `REVIEW.PRODUCT_ID`
- **Meaning:** Users can write reviews for multiple products, and each product can have reviews from multiple users. The REVIEW table stores feedback, ratings, and dates.

4.USERS<=>CUSTOMIZATIONS<=>PRODUCTS

- **Relationship:** Many-to-Many (resolved via CUSTOMIZATIONS table)
- **PK/FK Links:**
  - `USERS.USER_ID` → `CUSTOMIZATIONS.USER_ID`
  - `PRODUCTS.PRODUCT_ID` → `CUSTOMIZATIONS.PRODUCT_ID`
- **Meaning:** Users can create customizations for products. Products can be customized by multiple users. The CUSTOMIZATIONS table records the customization type and links it to both user and product.

5.PRODUCTS

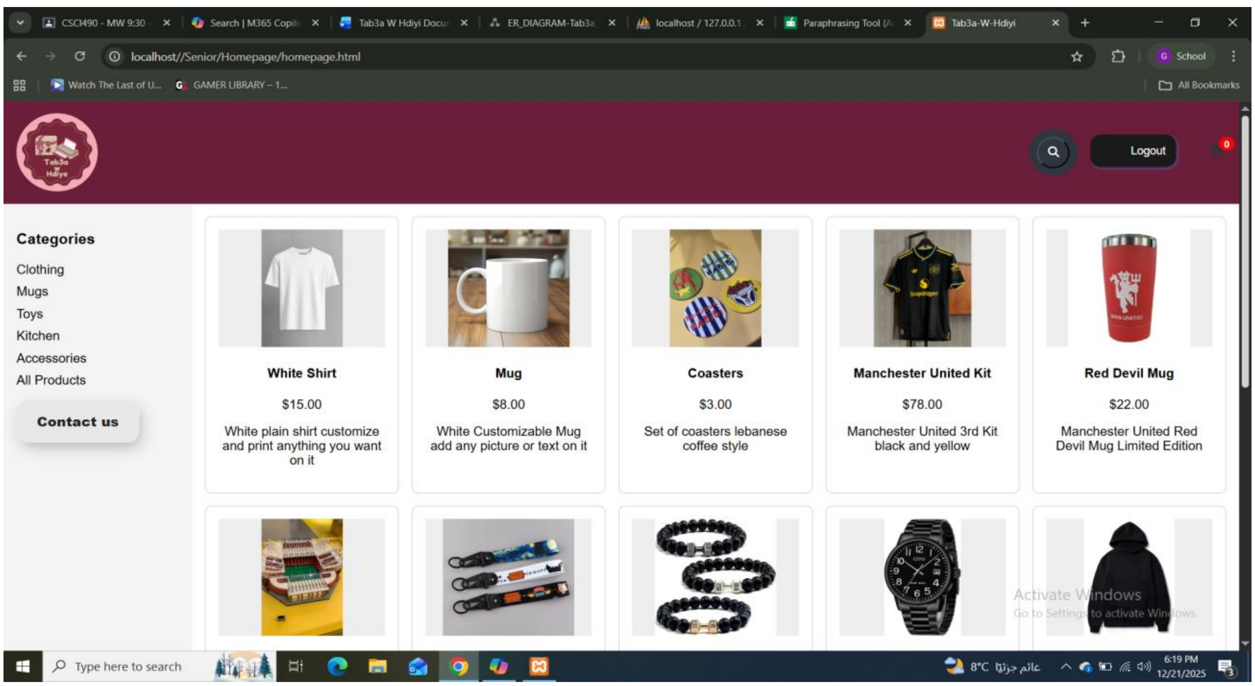
- Acts as a central entity connected to both **ORDERS** (via CONTAINS), **REVIEWS**, and **CUSTOMIZATIONS**.
- This design ensures that product data is consistent across sales, feedback, and personalization workflows



STORYBOARD:

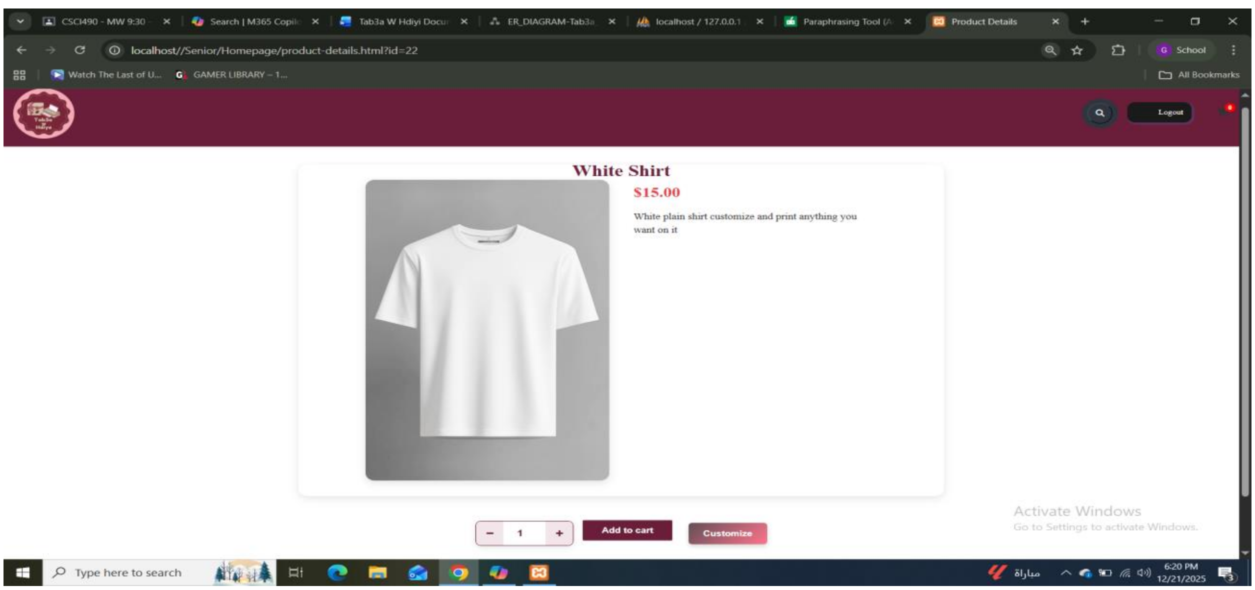
This section presents the visual flow and functional purpose of each major page in the Tab3a W Hdiyi e-commerce platform. It illustrates how users interact with the system from browsing products to customizing, purchasing, and managing orders.

🏠 HOMEPAGE:



- **Purpose:** Entry point for users to explore available products.
- **Features:**
  - Sidebar navigation with categories: Clothing, Mugs, Toys, Kitchen, Accessories, All Products.
  - Product grid showing image, name, price, and description.
  - “Contact Us” button for inquiries.
- **User Actions:** Browse products, filter by category, click to view details

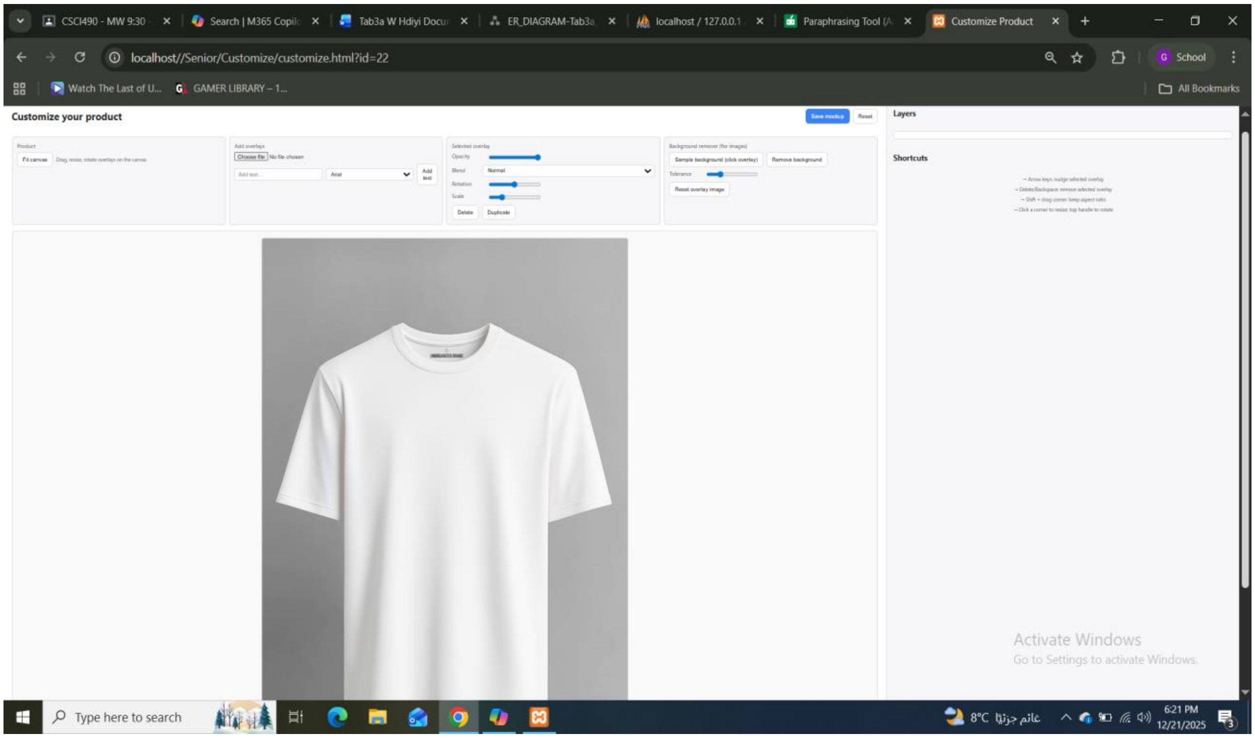
Product Details Page



- **Purpose:** Displays full information about a selected product.

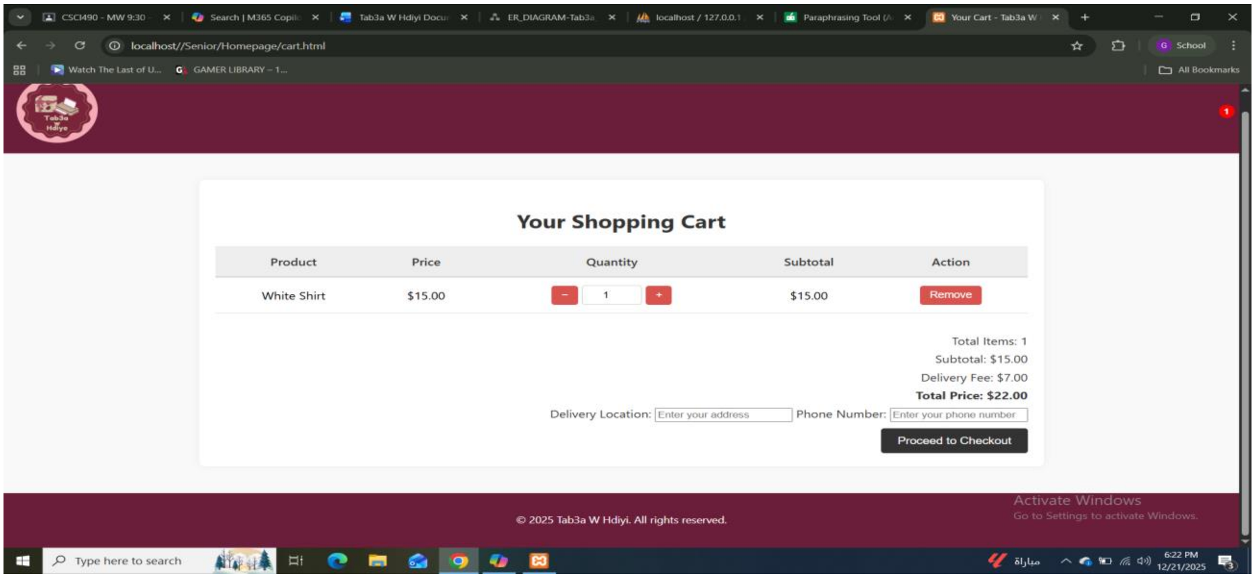
- **Features:**
  - Large product image and description.
  - Quantity selector and “Add to Cart” button.
  - “Customize” button for eligible products.
- **User Actions:** Add item to cart, initiate customization.

🎨 Customize Page



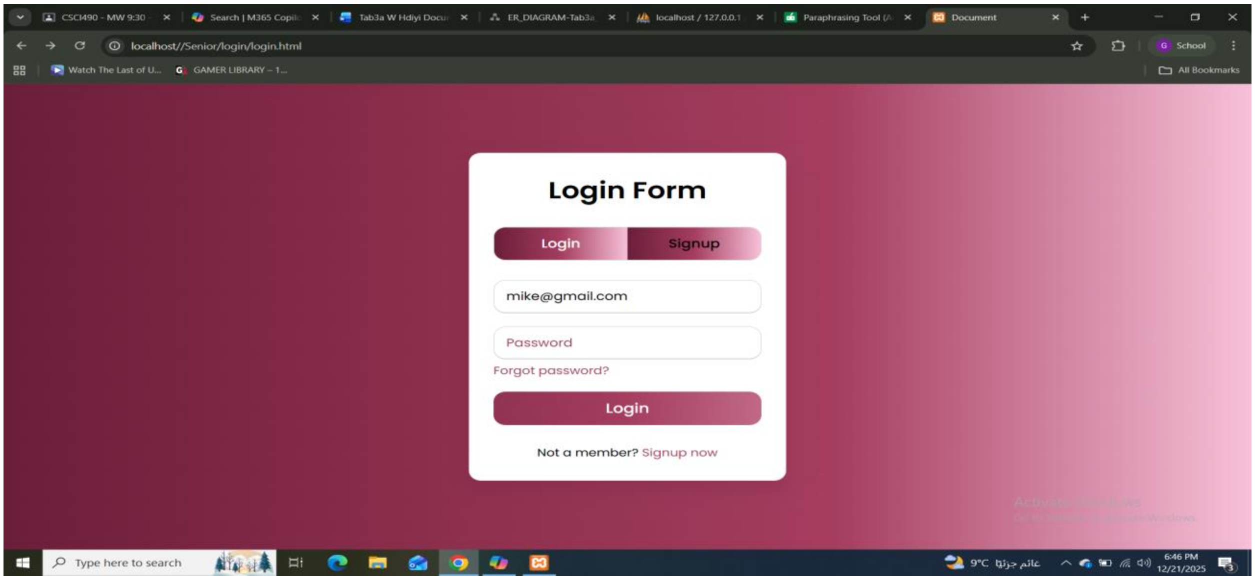
- **Purpose:** Allows users to personalize products with text or images.
- **Features:**
  - Canvas editor with overlay tools.
  - Text input, font selection, image upload.
  - Background remover, layer controls, opacity and rotation sliders.
- **User Actions:** Design product, preview changes, save customization.

🛒 Cart Page



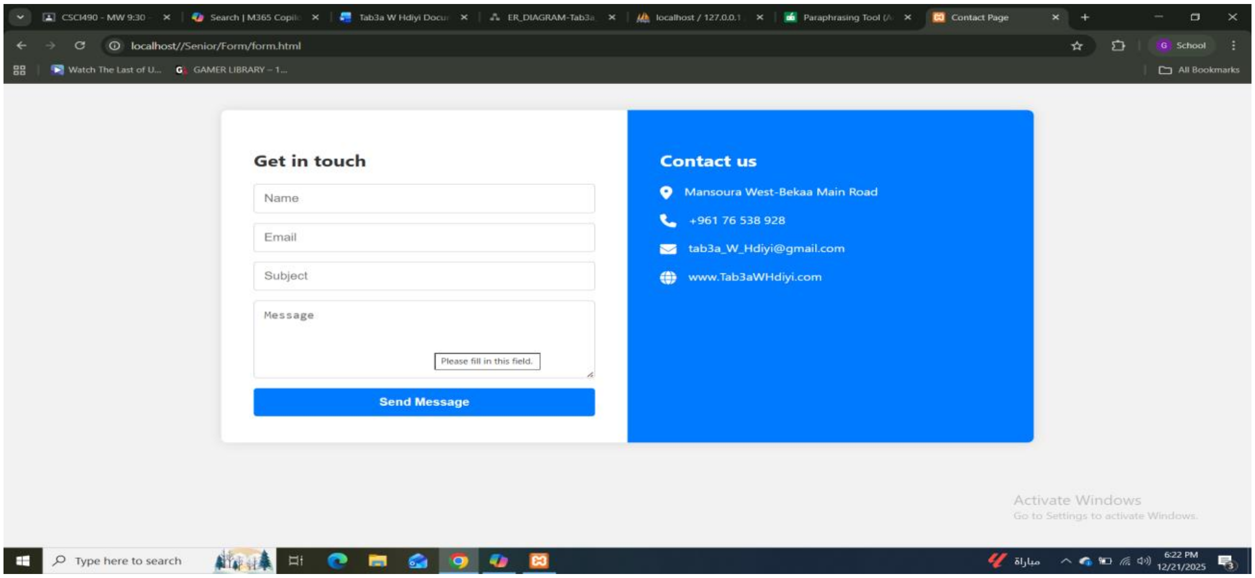
- **Purpose:** Summarizes selected items before checkout.
- **Features:**
    - Item list with price, quantity, subtotal, and remove option.
    - Delivery fee breakdown and total price.
    - Input fields for delivery location and phone number.
    - “Proceed to Checkout” button.
- **User Actions:** Review cart, enter delivery info, confirm purchase

🔐 Login Page



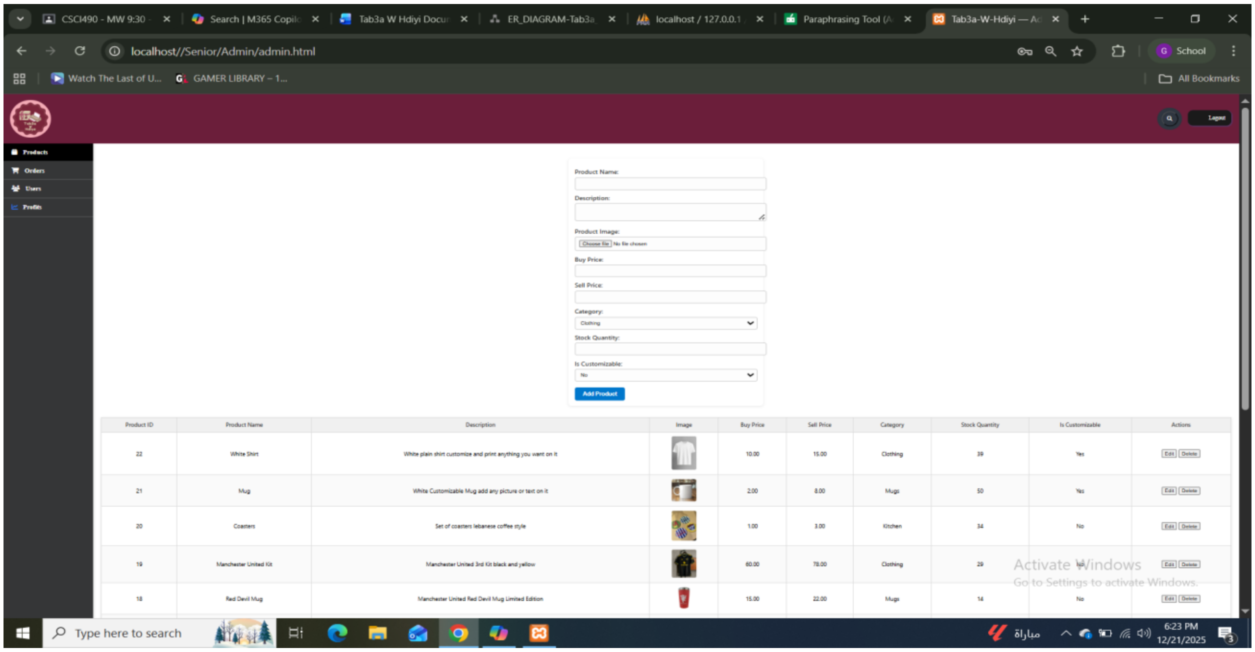
- **Purpose:** Authenticates users and grants access to personalized features.
- **Features:**
  - Email and password input fields.
  - “Forgot password?” link and “Signup” tab.
  - Login button and redirect to dashboard.
- **User Actions:** Log in, recover password, switch to signup.

📬 Contact Us Page



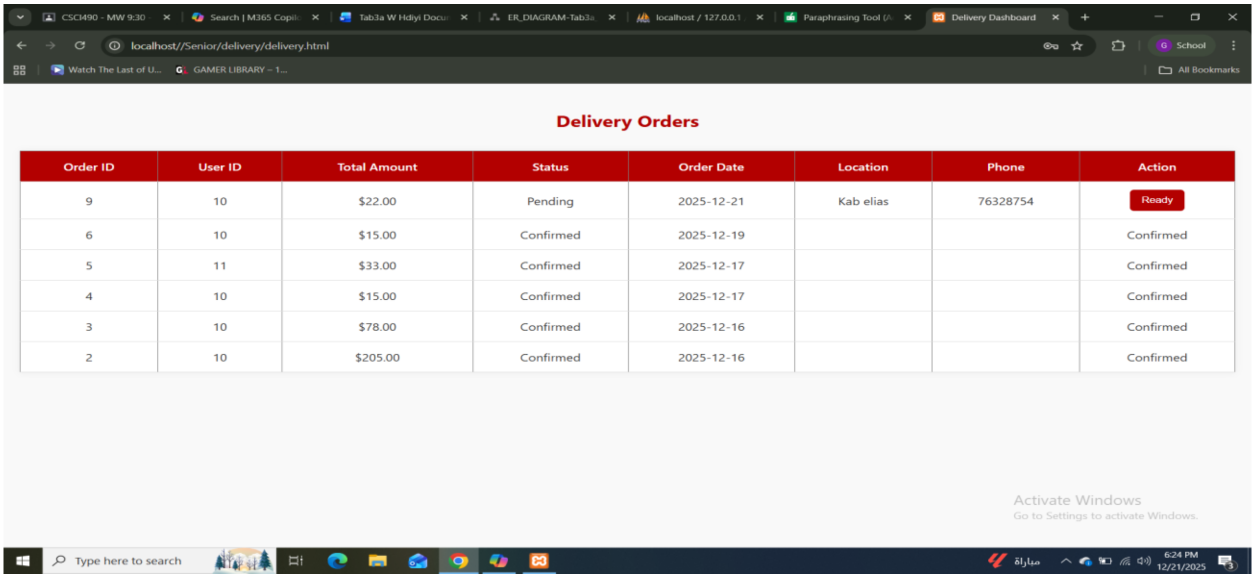
- **Purpose:** Enables users to reach out for support or inquiries.
- **Features:**
  - Form with fields: Name, Email, Subject, Message.
  - Contact details: Location, phone, email, website.
- **User Actions:** Submit message, view business contact info.

🛠️ Admin Dashboard



- **Purpose:** Backend interface for managing products.
- **Features:**
    - Add product form with fields for name, description, image, pricing, category, stock, and customizability.
    - Product table with edit/delete actions.
- **User Actions:** Add, edit, or delete products.

🚚 Delivery Orders Page



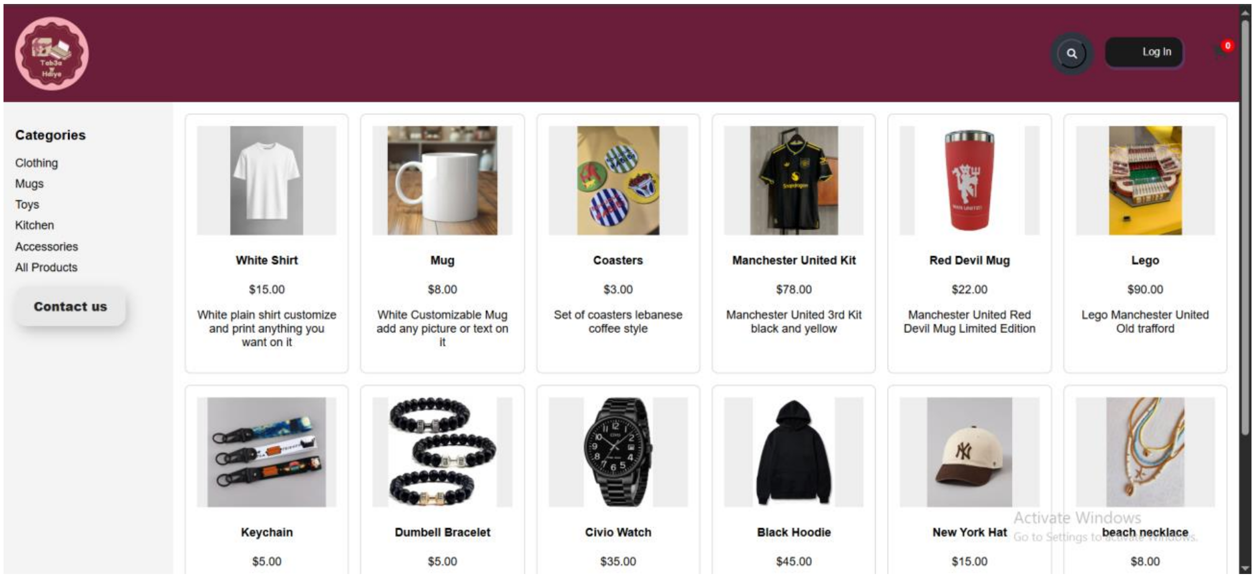
- **Purpose:** Tracks delivery status of confirmed orders.
- **Features:**
    - Table with columns: Order ID, User ID, Total Amount, Status, Order Date, Location, Phone, Action.
    - Status indicators: Pending, Confirmed, Ready.
- **User Actions:** View delivery queue, mark orders as ready.

Projects Parts

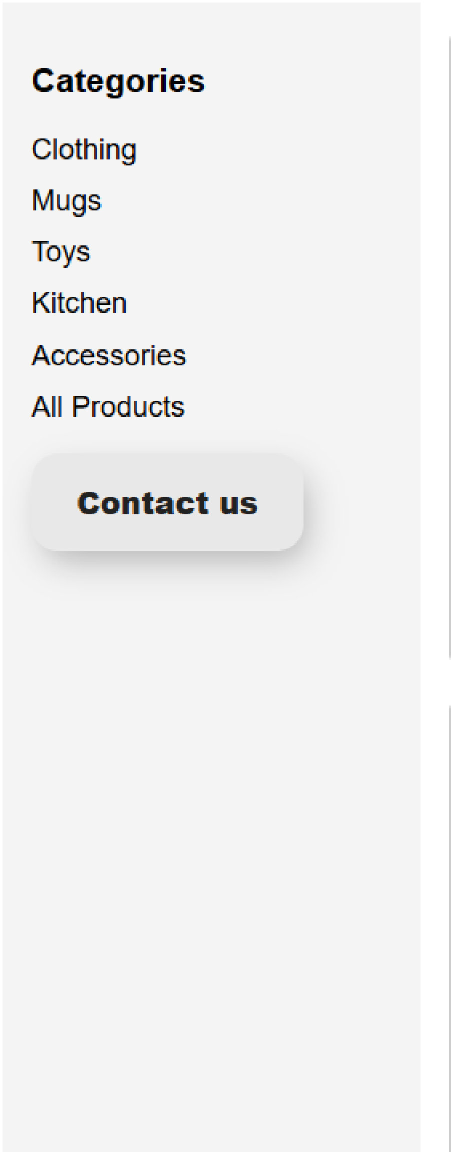
Homepage

Purpose:

The homepage is the starting point of the website. It shows all products and lets users explore categories.



Main Elements:



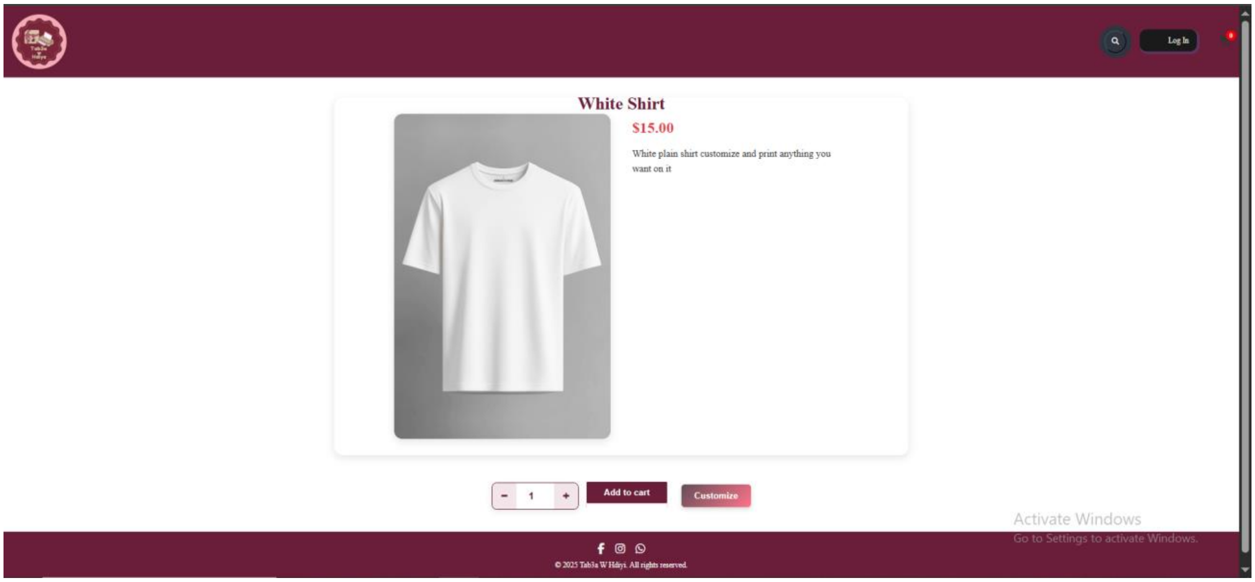
- **Category Buttons (Clothing, Mugs, Toys, Kitchen, Accessories, All Products):**
    - **Does:** Filters the product list to show only items from the chosen category.
    - **Should Do:** Instantly reload the product grid with matching items, without refreshing the whole page.
- **Contact Us Button:**
- **Does:** Redirects to the Contact Us page.
- **Should Do:** Provide easy access to support or inquiries.

- **Product Cards (Image, Name, Price, Description):**
  - **Does:** Clicking opens the product details page.
  - **Should Do:** Show the full product information and allow customization or purchase.

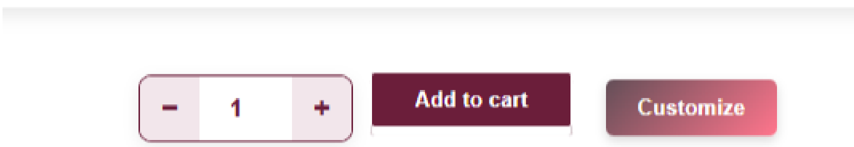
Product Details Page

Purpose:

Shows detailed information about one product and lets the user buy or customize it.



Main Elements:



- **Quantity Selector (+ / -):**
  - **Does:** Increases or decreases the number of items to buy.
  - **Should Do:** Prevent negative numbers and show updated subtotal.
- **Add to Cart Button:**
  - **Does:** Adds the product with the chosen quantity to the cart.
  - **Should Do:** Confirm with a message like "Item added to cart" and update the cart icon.
- **Customize Button:**
  - **Does:** Opens the Customize Page if the product is customizable.
  - **Should Do:** Pass product details to the customization tool.

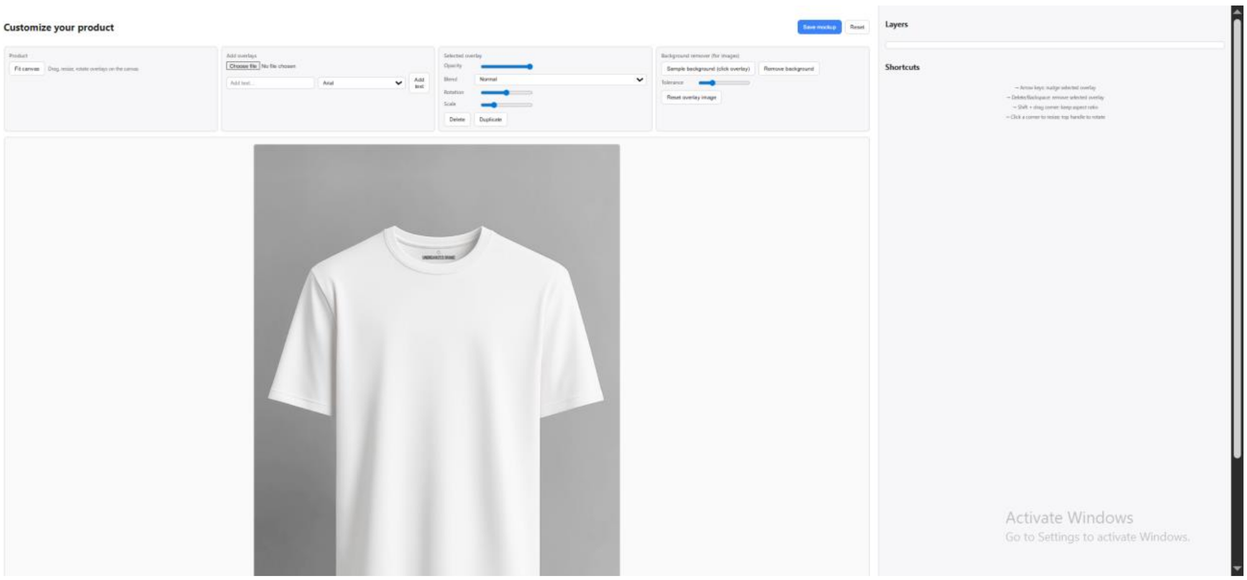
- **Header Buttons:**

- **Search Icon:** Opens a search bar to find products.
- **Cart Icon:** Redirects to the Cart Page.
- **Logout Button:** Logs the user out and returns to the Login Page.

Customize Page

Purpose:

Allows users to personalize products with text or images.



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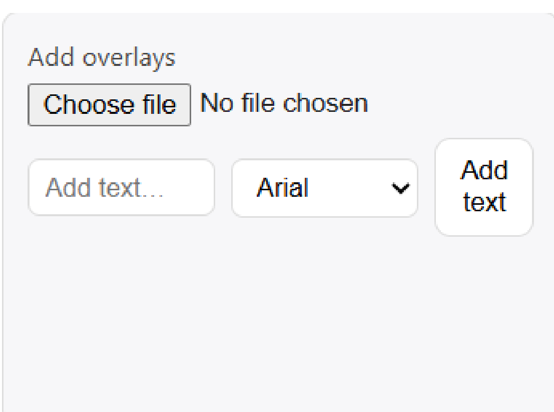
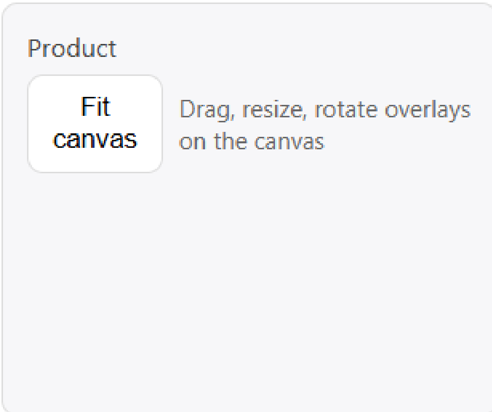
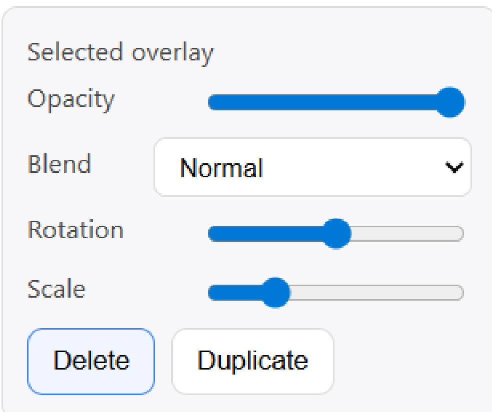
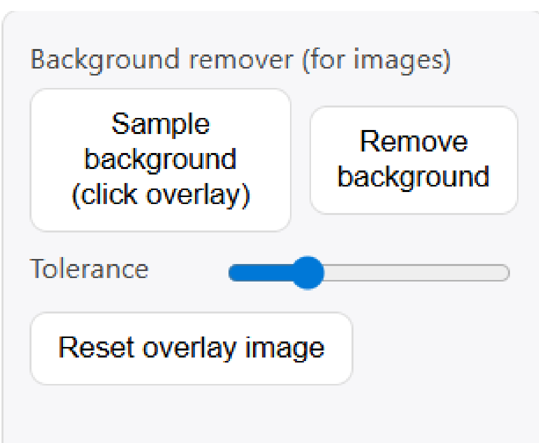








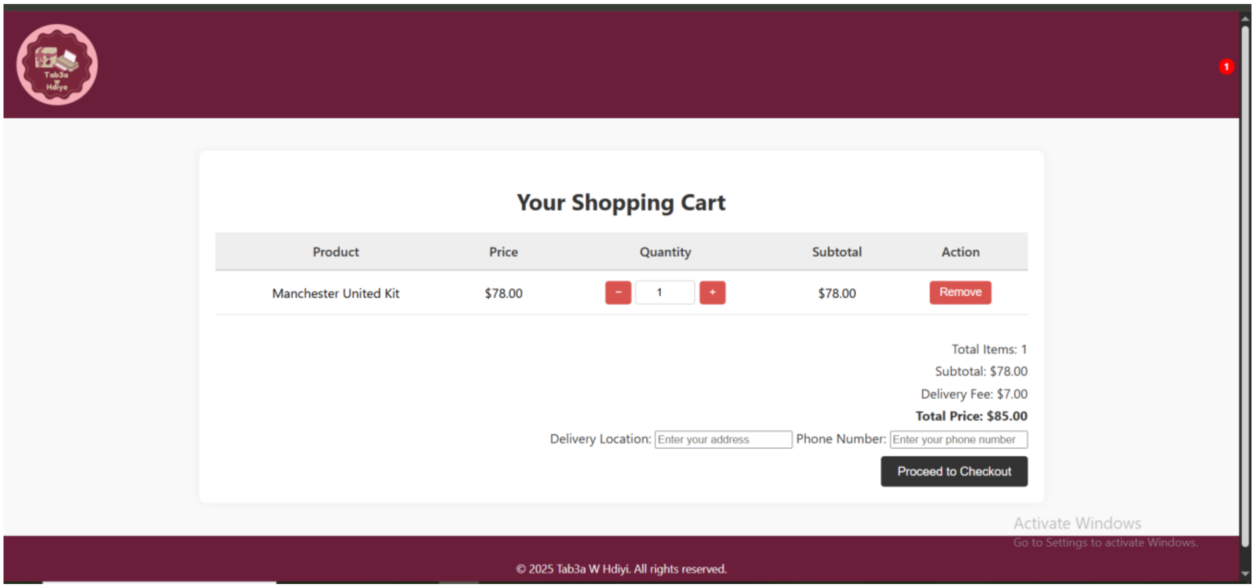
- **Fit Canvas Button:**
  - **Does:** Resizes the product image to fit the canvas.
  - **Should Do:** Reset overlays to proper alignment.
- **Choose File (Image Upload):**

    - **Does:** Uploads an image overlay.
    - **Should Do:** Validate file type (e.g., JPG, PNG).
- **Add Text Button:**
    - **Does:** Adds custom text overlay.
    - **Should Do:** Allow font, size, and color changes.
- **Font Selector:**
    - **Does:** Changes text style.
    - **Should Do:** Preview instantly on canvas.
- **Opacity Slider:**
    - **Does:** Adjusts transparency of overlays.
    - **Should Do:** Range from 0% (invisible) to 100% (solid).
- **Rotation & Scale Controls:**
    - **Does:** Rotates or resizes overlays.
    - **Should Do:** Keep proportions when resizing.
- **Delete/Duplicate Overlay Buttons:**
    - **Does:** Removes or copies overlays.
    - **Should Do:** Confirm deletion to avoid mistakes.
- **Background Remover Tools:**
    - **Does:** Removes background from uploaded images.
    - **Should Do:** Allow tolerance adjustment for cleaner results.
- **Layers Panel:**
- **Does:** Manages multiple overlays.
- **Should Do:** Let users reorder overlays easily.

Cart Page

Purpose:

Shows selected items before checkout.



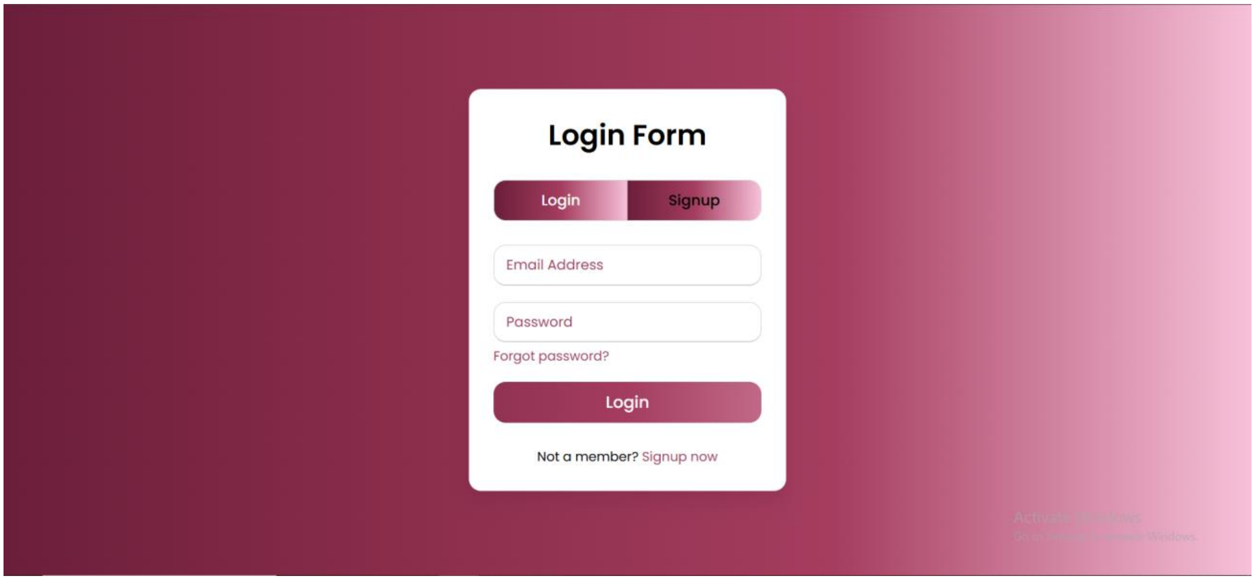
Main Elements:

- **Remove Button:**
    - **Does:** Deletes product from cart.
    - **Should Do:** Update totals immediately.
- **Delivery Location Input:**
    - **Does:** Collects address.
    - **Should Do:** Validate that it is not empty.
- **Phone Number Input:**
    - **Does:** Collects contact number.
    - **Should Do:** Validate correct format (digits only).
- **Proceed to Checkout Button:**
- **Does:** Confirms order and moves to checkout.
- **Should Do:** Save order details and show confirmation page

Login Page

Purpose:

Authenticates users to access their accounts.



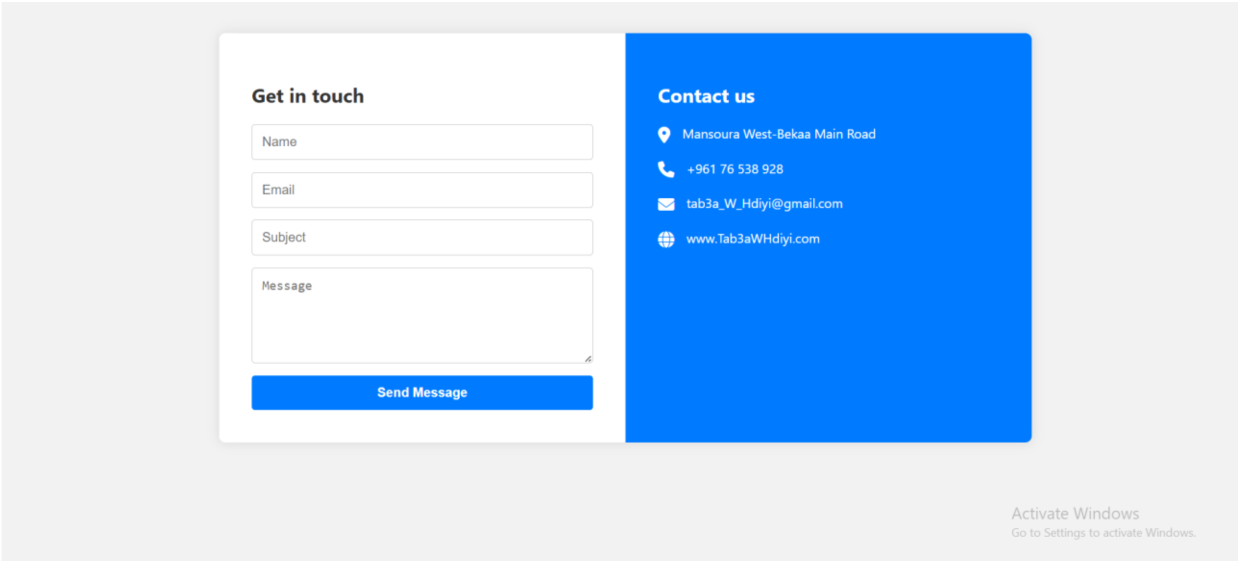
Main Elements:

- **Email Input:**
  - **Does:** Collects user email.
  - **Should Do:** Validate format (e.g., name@email.com).
- **Password Input:**
  - **Does:** Collects password.
  - **Should Do:** Hide characters for security.
- **Login Button:**
  - **Does:** Authenticates credentials.
  - **Should Do:** Redirect to dashboard if correct, show error if wrong.
- **Forgot Password Link:**
  - **Does:** Opens password recovery.
  - **Should Do:** Send reset email.
- **Signup Tab/Button:**
- **Does:** Switches to signup form.
- **Should Do:** Allow new account creation.

Contact Us Page

Purpose:

Lets users send inquiries or feedback.



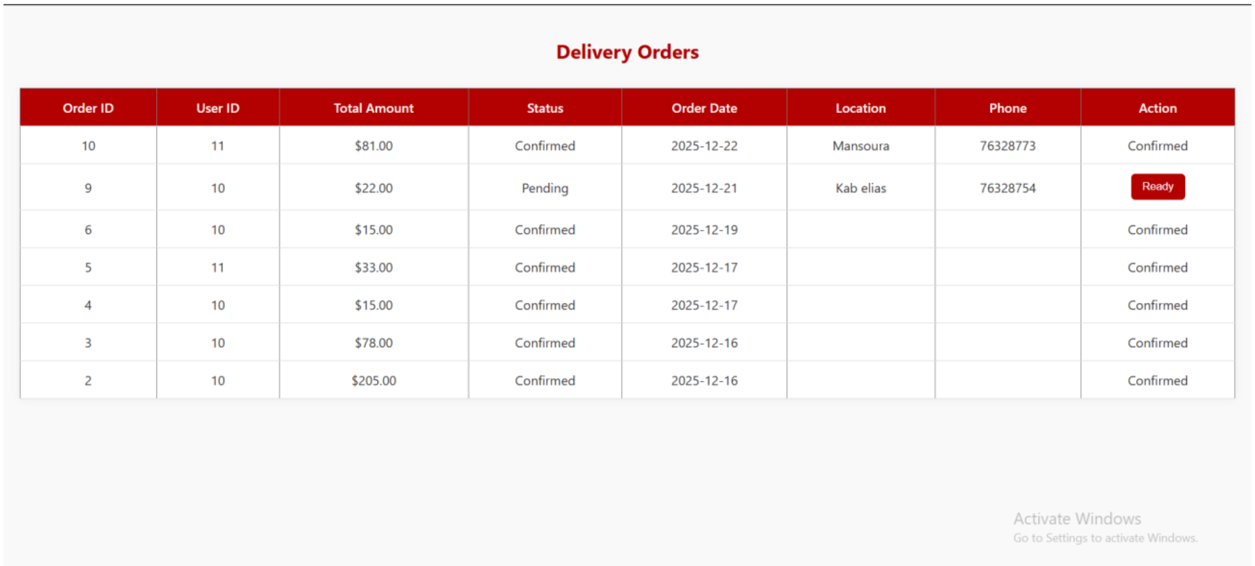
Main Elements:

- **Form Fields (Name, Email, Subject, Message):**
    - **Does:** Collects user input.
    - **Should Do:** Validate required fields before submission.
- **Send Message Button:**
    - **Does:** Sends inquiry.
    - **Should Do:** Show success message and email admin.
- **Business Contact Info:**
- **Does:** Displays static info.
- **Should Do:** Always be visible for quick reference.

Delivery Orders Page

Purpose:

Tracks delivery status of orders.



Delivery Orders

Order ID	User ID	Total Amount	Status	Order Date	Location	Phone	Action
10	11	$81.00	Confirmed	2025-12-22	Mansoura	76328773	Confirmed
9	10	$22.00	Pending	2025-12-21	Kab elias	76328754	Ready
6	10	$15.00	Confirmed	2025-12-19			Confirmed
5	11	$33.00	Confirmed	2025-12-17			Confirmed
4	10	$15.00	Confirmed	2025-12-17			Confirmed
3	10	$78.00	Confirmed	2025-12-16			Confirmed
2	10	$205.00	Confirmed	2025-12-16			Confirmed

Main Elements:

- **Order Table:**
    - **Does:** Displays order details.
    - **Should Do:** Update dynamically when status changes.
- **Action Column (Ready / Confirmed):**
    - **Does:** Updates order status.
    - **Should Do:** Notify user when status changes.
- **Status Field:**
- **Does:** Shows current order state.
- **Should Do:** Only allow valid transitions (Pending → Confirmed → Ready)

Admin Dashboard

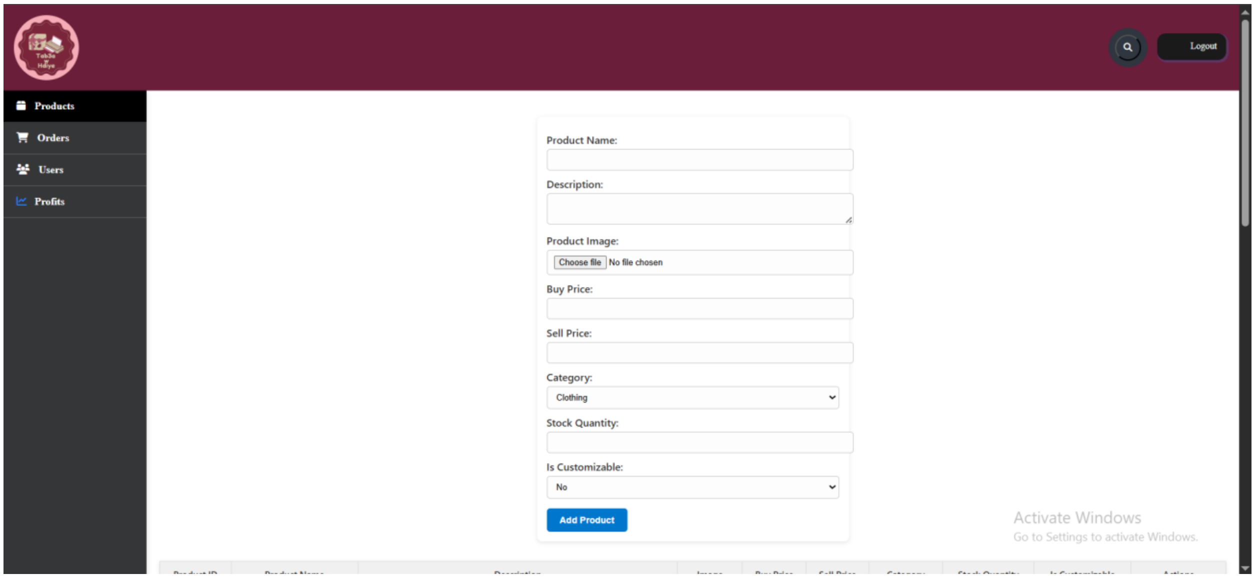
The Admin Dashboard is the control center for managing the Tab3a W Hdiyi platform. It includes four main tabs: **Products**, **Orders**, **Users**, and **Profits**. Each tab has its own purpose and tools to help the admin manage the system smoothly.

Products Tab

Purpose:

To add, edit, and delete products in the store.

Add Product Form



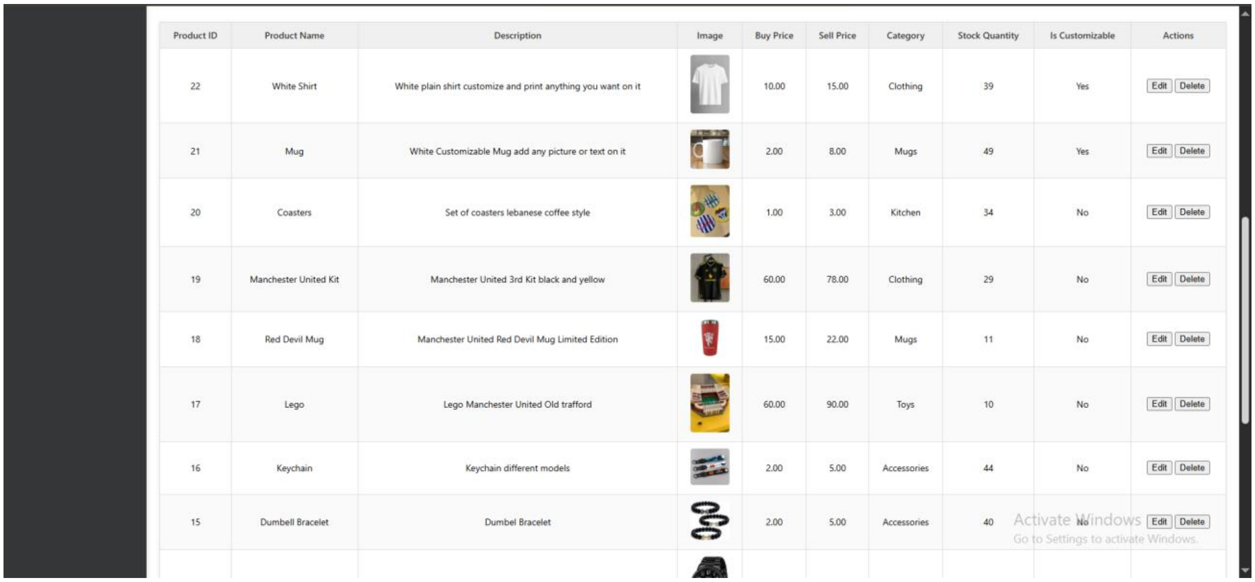
Fields and Buttons:

- **Product Name:**
    - Admin types the name of the product.
    - *Should check* that it's not empty.
- **Description:**
    - Explains what the product is.
    - *Should help* customers understand the item.
- **Product Image (Choose File):**
    - Uploads a photo.
    - *Should accept* only image files and show a preview.
- **Buy Price:**
    - Cost paid by admin.
    - *Should be* a number greater than zero.
- **Sell Price:**
    - Price for customers.

- *Should be* higher than buy price.

- **Category (Dropdown):**
    - Selects where the product belongs (Clothing, Mugs, etc.).
    - *Should organize* products on the homepage.
- **Stock Quantity:**
    - Number of items available.
    - *Should prevent* negative numbers.
- **Is Customizable (Yes/No):**
    - Marks if the product can be personalized.
    - *Should control* whether the “Customize” button appears.
- **Add Product Button:**
- Saves the product.
- *Should validate* all fields and refresh the product list.

Product Table



Product ID	Product Name	Description	Image	Buy Price	Sell Price	Category	Stock Quantity	Is Customizable	Actions
22	White Shirt	White plain shirt customize and print anything you want on it		10.00	15.00	Clothing	39	Yes	Edit Delete
21	Mug	White Customizable Mug add any picture or text on it		2.00	8.00	Mugs	49	Yes	Edit Delete
20	Coasters	Set of coasters lebanese coffee style		1.00	3.00	Kitchen	34	No	Edit Delete
19	Manchester United Kit	Manchester United 3rd Kit black and yellow		60.00	78.00	Clothing	29	No	Edit Delete
18	Red Devil Mug	Manchester United Red Devil Mug Limited Edition		15.00	22.00	Mugs	11	No	Edit Delete
17	Lego	Lego Manchester United Old trafford		60.00	90.00	Toys	10	No	Edit Delete
16	Keychain	Keychain different models		2.00	5.00	Accessories	44	No	Edit Delete
15	Dumbell Bracelet	Dumbel Bracelet		2.00	5.00	Accessories	40	No	Edit Delete

Columns:

- **Product ID:** Unique number.
- **Product Name & Description:** Basic info.
- **Image:** Thumbnail preview.

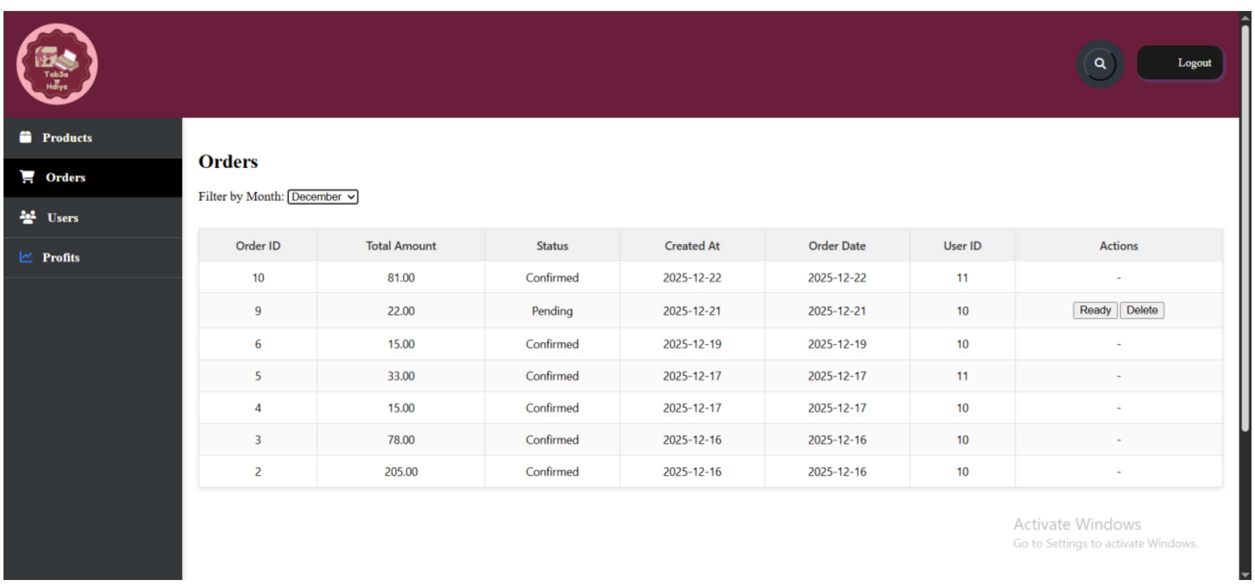
- **Buy/Sell Price:** Cost and selling price.
- **Category:** Product group.
- **Stock Quantity:** Inventory count.
- **Is Customizable:** Yes or No.
- **Actions:**
- **Edit Button:**
  - Opens product in form.
  - *Should load* all fields for editing.
- **Delete Button:**
- Removes product.
- *Should ask* for confirmation before deleting

Orders Tab

Purpose:

To view and manage customer orders.

Orders Table



Columns:

- **Order ID:** Unique number.
- **Total Amount:** Full price of the order.
- **Status:**
    - Shows if the order is Pending, Confirmed, or Ready.
    - *Should update* based on delivery progress.
- **Created At / Order Date:**
    - When the order was made.
- **User ID:**
    - Links order to the customer.
- **Actions:**
- **Ready Button:**
    - Marks order as ready for delivery.
    - *Should change* status to “Ready”.
- **Delete Button:**
- Removes the order.
- *Should confirm* before deleting.

Extra Feature:

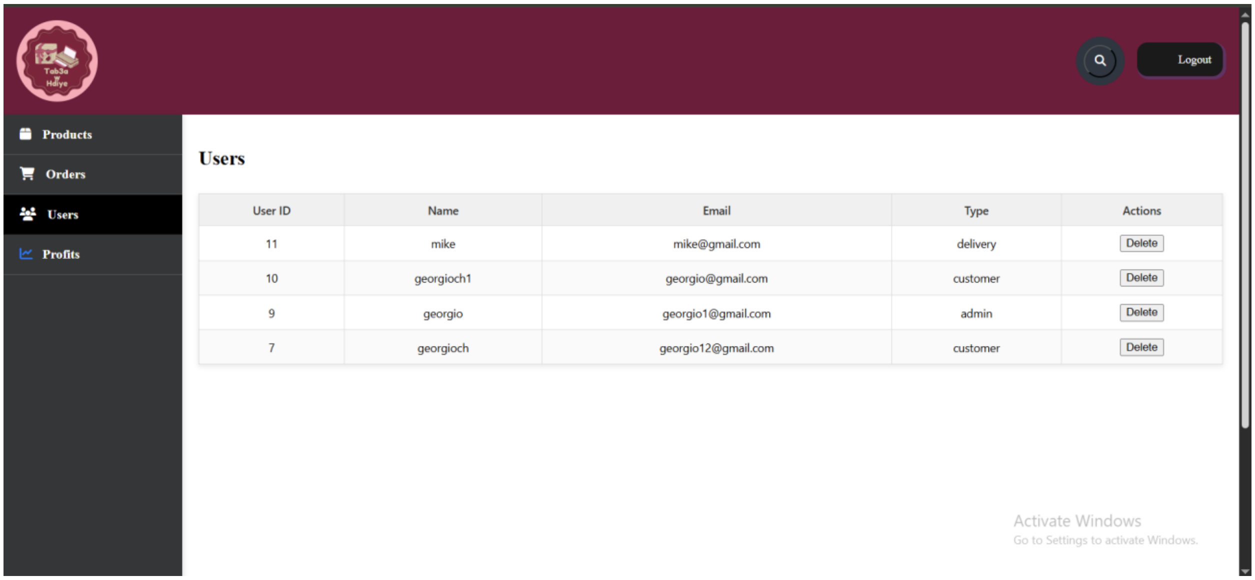
- **Filter by Month (Dropdown):**
- Filters orders by selected month.
- *Should reload* table with matching results.

Users Tab

Purpose:

To manage all registered users.

Users Table



Columns:

- **User ID:** Unique number.
- **Name:** Username.
- **Email:** Contact email.
- **Type:**
  - Shows if the user is an Admin, Customer, or Delivery person.
  - *Should control* what they can access.
- **Actions:**
- **Delete Button:**
- Removes the user.
- *Should confirm* before deleting.

Extra Features:

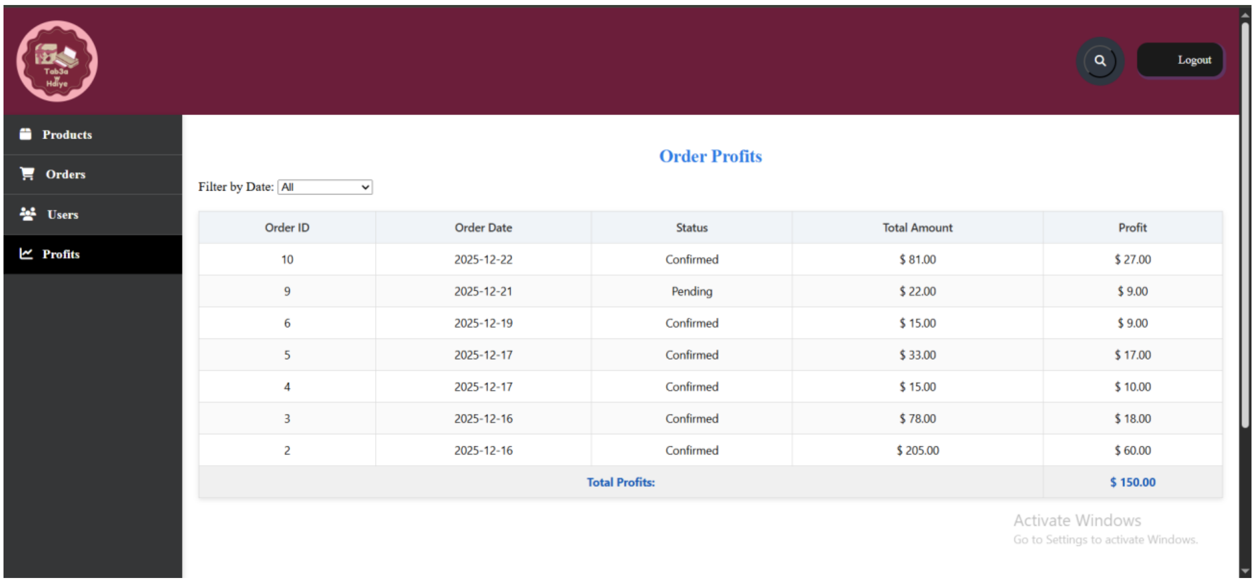
- **Search Icon (Top Right):**
  - Finds users by name or email.
  - *Should filter* the table instantly.
- **Logout Button:**
- Logs out the admin.
- *Should redirect* to Login Page.

Profits Tab

Purpose:

To track profits from confirmed orders.

Profits Table



Columns:

- **Order ID:** Unique number.
- **Order Date:** When the order was placed.
- **Status:**
  - Only confirmed orders count toward profit.
- **Total Amount:** Full price of the order.
- **Profit:**
- Difference between sell price and buy price.
- *Should be calculated* automatically.

Extra Feature:

- **Filter by Date (Dropdown):**
- Filters orders by time period.

- *Should update* the table and total profit.

Summary:

- **Total Profits (Bottom of Table):**
- Shows the sum of all profits.
- *Should update* when filters change

Future Improvements

The current Tab3a W Hdiyi system is a web-based platform that allows customers to browse products, customize items, place orders, and track deliveries. While the system is functional and covers the basic needs of an e-commerce store, there are many opportunities to expand and improve it in the future. These improvements will make the platform more user-friendly, more secure, and more scalable, while also opening the door to new business opportunities.

📱 Mobile Application Development

One of the biggest improvements planned for the future is to build a **mobile application** for both Android and iOS. Many customers prefer shopping through mobile apps because they are faster, easier to access, and can send notifications directly to the phone. The app would allow users to browse products, customize items with touch gestures, and place orders with just a few taps. It would also include push notifications to alert customers when their order status changes (for example, when it is confirmed or ready for delivery). Another feature could be an offline mode, where customers can still view saved products even without internet access. This mobile app would make the platform more modern and competitive compared to other online stores.

💳 Payment Gateway Integration

Currently, the system relies on cash-on-delivery. In the future, it should support **online payment gateways** such as PayPal, Stripe, or local mobile wallets. This would allow customers to pay securely with their credit cards or digital wallets. Online payments reduce the risk of failed deliveries and increase customer trust. The system should also generate receipts automatically and send them to the customer's email. Adding payment gateways will make the platform more professional and suitable for larger audiences.

🌍 Multi-Language Support

Since Tab3a W Hdiyi is designed for a wide audience, adding **multi-language support** will make it more accessible. For example, the platform could support Arabic, French, and English. Customers would be able to choose their preferred language from a dropdown menu, and the entire interface would change accordingly. Admins could manage translations through a dashboard, ensuring that product descriptions and messages are consistent across languages. This improvement would expand the customer base internationally.

📦 Customer Order Tracking

Another important improvement is to allow customers to **track their orders** directly from their account. Instead of waiting for a phone call, customers could log in and see the status of their order: Pending, Confirmed, Ready, or Delivered. Delivery staff could update the status from their own app or dashboard, and customers would receive real-time notifications. This feature would improve transparency and customer satisfaction.

📊 Advanced Analytics Dashboard

For admins, a **business analytics dashboard** would be very useful. It could show charts and graphs about sales, profits, and top-selling products. Admins could filter data by day, week, or month, and export reports as PDF or Excel files. This would help in making better business decisions, such as which products to promote or which categories are most profitable. Analytics would also highlight customer trends, such as which times of year have the highest sales.

🔒 Enhanced Security Features

Security is always important in online systems. Future improvements should include stronger password rules, such as requiring uppercase letters, numbers, and special characters. Two-factor authentication could be added, where users must confirm their login with a code sent to their phone or email. File uploads should be scanned to prevent harmful files, and all data should be encrypted. Role-based access control should also be improved, so that admins, delivery staff, and customers only see the parts of the system they are allowed to access.

🎨 Improved Customization Tools

The customization page is already a strong feature, but it can be made even better. Future improvements could include more fonts, colors, and design templates. Customers could preview their customized product in **3D**, rotating it to see how it looks from different angles. AI-assisted background removal could make image uploads cleaner and faster. These improvements would make customization more fun and professional.

🤝 Social Media Integration

To increase marketing reach, the platform could be connected to **social media platforms** like Instagram and Facebook. Customers could share their customized products directly from the website or app. This would create free advertising through user-generated content. Admins could also run promotions where customers get discounts if they share their product designs online.

🛠️ Scalability and Cloud Hosting

Currently, the system runs on a local server. In the future, it should be hosted on the **cloud** (such as AWS or Microsoft Azure). Cloud hosting would allow the system to handle more users and orders at the same time. It would also provide automatic backups, disaster recovery, and better performance. This improvement would make the platform ready for large-scale use.

🧑‍🤝‍🧑 Community Features

Adding **community features** would make the platform more interactive. Customers could leave reviews and ratings for products, helping others decide what to buy. A loyalty points system could reward repeat buyers with discounts. Customers could also suggest new product ideas, creating a sense of community and engagement. These features would build stronger relationships between the business and its customers.

📧 Email and SMS Notifications

Future versions of the system could send **automatic notifications** to customers. For example, when an order is confirmed, the customer could receive an email or SMS message. Notifications could also remind customers of promotions or new products. This would keep customers engaged and informed.

🛍️ Personalized Recommendations

Using customer data, the system could suggest products based on past purchases. For example, if a customer buys mugs, the system could recommend coasters or keychains. Personalized recommendations would increase sales and improve the shopping experience.

🌐 Integration with Delivery Services

Instead of managing delivery manually, the system could connect with local delivery companies. Orders could be sent directly to the delivery service, and tracking numbers could be shared with customers. This would make the delivery process faster and more reliable.

Summary

By adding these improvements, Tab3a W Hdiyi can grow from a simple web platform into a **full-featured e-commerce ecosystem**. The mobile app will make it more accessible, payment gateways will make it more professional, and analytics will make it smarter. Security, customization, and community features will ensure that both customers and admins enjoy using the system. These future plans show that the project is not just a student exercise, but a foundation for a real business.

System Architecture

The Tab3a W Hdiyi platform is built using a **three-layer architecture**: the client layer, the server layer, and the database layer. Each layer has its own responsibilities, and together they form a complete e-commerce system.

1. **Client Layer (Frontend)**

This is the part of the system that customers and admins see and interact with. It is built using **HTML, CSS, and JavaScript**.

- **HTML** provides the structure of the pages (login form, product grid, cart table, etc.).
- **CSS** controls the design, colors, and layout. For example, the gradient pink background on the login page and the card-style containers are CSS features.
- **JavaScript** adds interactivity. It updates totals in the cart, validates forms before submission, and controls dynamic elements like the customization canvas.

The client layer communicates with the server by sending requests (for example, when a customer clicks "Add to Cart" or when an admin clicks "Add Product").

2. **Server Layer (Backend)**

The backend is built using **PHP** and handles all the business logic. It acts as the "brain" of the system.

- **Authentication:** When a user logs in, PHP checks the email and password against the database. If valid, it creates a session and redirects the user to the correct dashboard.
- **Product Management:** Admin actions like adding, editing, or deleting products are processed here. PHP validates the input and updates the database.
- **Order Handling:** When a customer places an order, PHP saves the order details, calculates totals, and updates the order status.
- **Customization Saving:** When a customer customizes a product, PHP stores the design details (text, images, overlays) in the database.
- **Security:** PHP sanitizes inputs to prevent SQL injection and ensures that only authorized users can access admin features.

3. **Database Layer**

The database is built using **MySQL**. It stores all the information needed by the system.

- **Users Table:** Stores user details (ID, name, email, password, role).
- **Products Table:** Stores product details (ID, name, description, image path, buy/sell price, category, stock, customizable flag).
- **Orders Table:** Stores order details (ID, user ID, total amount, status, dates).
- **Profits Table:** Calculates profit for each order based on buy and sell prices.
- **Customizations Table (optional):** Stores design details for customized products.

The database ensures data consistency. For example, when stock decreases after an order, the change is reflected immediately in the Products table.

4. **Data Flow**

Here's how the layers communicate:

1. **Customer Action:** A customer clicks "Add to Cart" on the frontend.
2. **Server Processing:** PHP receives the request, validates it, and updates the Orders table in MySQL.
3. **Database Update:** MySQL saves the new order and adjusts stock.
4. **Response:** PHP sends a confirmation back to the frontend, which shows "Item added to cart."

This flow happens for every major action: login, signup, customization, checkout, admin product management, and delivery updates.

5. **Admin Dashboard Architecture**

The Admin Dashboard follows the same three-layer model but with extra privileges.

- **Frontend:** Admin sees forms and tables for Products, Orders, Users, and Profits.
- **Backend:** PHP checks that the logged-in user is an admin before allowing access.
- **Database:** Admin actions directly update the Products, Orders, Users, and Profits tables.

6. **Scalability Considerations**

Although the system currently runs on localhost, the architecture is designed to scale. In the future:

- The client layer can be rebuilt with **React** for faster updates.
- The server layer can move to **Node.js or .NET** for higher performance.

- The database can be hosted on the **cloud** with replication and backups.

This layered design makes it easier to add new features without breaking the existing system.

Accessibility Features

Accessibility means making sure that **everyone can use the Tab3a W Hdiyi platform**, including people with disabilities or those using different devices. Adding accessibility features is important because it makes the system more inclusive, professional, and user-friendly. Below are the key accessibility features that can be included or improved in the system.

1. **Clear Labels and Instructions**

- Every input field (like email, password, product name, or delivery address) should have a clear label.
- Labels help screen readers announce the purpose of the field to visually impaired users.
- Example: Instead of just showing a blank box, the system should say “Enter your email” or “Enter your password.”

2. **Keyboard Navigation**

- Users should be able to move through the system using only the keyboard (Tab, Enter, Arrow keys).
- Buttons like “Login,” “Add to Cart,” or “Checkout” should be reachable without a mouse.
- This helps users with motor disabilities or those who prefer keyboard shortcuts.

3. **Color Contrast**

- Text should be easy to read against the background.

- For example, the pink gradient background on the login page should have dark text so it is readable.
- High contrast ensures that users with low vision can still see the content clearly.

4. **Alternative Text for Images**

- Every product image should have an **alt text** description.
- Example: A mug image should have alt text like “White customizable mug.”
- Screen readers use alt text to describe images to blind users.

5. **Responsive Design**

- The system should adapt to different screen sizes (desktop, tablet, mobile).
- Buttons and text should resize automatically.
- This helps users who rely on mobile devices or larger text settings.

6. **Form Validation with Feedback**

- When users make a mistake (like leaving a field empty or typing an invalid phone number), the system should give clear feedback.
- Example: “Please enter a valid phone number” instead of just showing an error code.
- Feedback should be both visual (red text) and audible for screen readers.

7. **Accessible Buttons**

- Buttons should be large enough to click easily.
- They should have clear text labels instead of only icons.
- Example: A “Login” button should say “Login” instead of just showing a lock icon.

8. **Language Options**

- Accessibility also means supporting different languages.
- Users should be able to switch between Arabic, English, and French.
- This helps people who are not fluent in one language.

9. **Screen Reader Compatibility**

- The system should be tested with screen readers like NVDA or JAWS.
- This ensures that all text, buttons, and links are announced properly.
- Example: When a user clicks "Add to Cart," the screen reader should say "Item added to cart."

10. **Error Prevention**

- Prevent users from making mistakes that are hard to fix.
- Example: Confirm before deleting a product in the admin dashboard.
- This helps users with cognitive disabilities who may click buttons by mistake.

11. **Accessible Customization Tools**

- The customization page should allow keyboard shortcuts for adding text or images.
- Sliders for opacity and rotation should also have text input options.
- This ensures that users who cannot use a mouse can still customize products.

12. **Consistent Layout**

- Pages should follow a consistent design so users don't get confused.
- Example: Login, Signup, and Dashboard should all use the same style of buttons and forms.
- Consistency helps users with learning difficulties.

Summary

By adding accessibility features, Tab3a W Hdiyi becomes more inclusive and professional. Clear labels, keyboard navigation, color contrast, alt text, and responsive design make the system usable for everyone. Accessibility is not just about helping people with disabilities—it also improves the experience for all users, making the platform easier, faster, and more enjoyable to use.

Technology Stack & Tools Used

The Tab3a W Hdiyi platform is built using a combination of **frontend, backend, and database technologies**, along with supporting tools for development and testing. Each part of the stack was chosen to balance simplicity, functionality, and scalability.

🖥️ Frontend (Client-Side)

The frontend is the part of the system that customers and admins interact with directly. It is responsible for the design, layout, and user experience.

- **HTML (HyperText Markup Language):**
- Provides the structure of all pages, including forms, product grids, and tables.
- Example: The login form and product cards are built with HTML elements.
- **CSS (Cascading Style Sheets):**

Controls the visual design, colors, and layout.
Example: The gradient backgrounds, button styles, and responsive layouts are CSS features.

- **JavaScript:**

Adds interactivity and dynamic behavior.
Example: Updating cart totals instantly, validating forms before submission, and controlling the customization canvas.

Together, these technologies ensure that the platform is visually appealing, responsive, and easy to use.

⚙️ Backend (Server-Side)

The backend is the "brain" of the system. It processes requests, applies business logic, and communicates with the database.

- **PHP (Hypertext Preprocessor):**

Handles authentication, product management, order processing, and customization saving.
Example: When a customer logs in, PHP checks their credentials against the database and creates a session.

- **Session Management:**

Ensures that each user (customer, admin, delivery staff) only sees their own data.
Example: Customers see only their orders, while admins see all orders.

- **Security Logic:**

PHP sanitizes inputs to prevent SQL injection and ensures that only authorized users can access admin features.

🗄️ Database Layer

The database stores all the information needed by the system. It is built using **MySQL**, a reliable relational database.

- **Users Table:** Stores user details (ID, name, email, password, role).
- **Products Table:** Stores product details (ID, name, description, image path, buy/sell price, category, stock, customizable flag).
- **Orders Table:** Stores order details (ID, user ID, total amount, status, dates).
- **Profits Table:** Calculates profit for each order.
- **Customizations Table (optional):** Stores design details for customized products.

MySQL ensures data consistency and supports queries for analytics, such as calculating total profits or filtering orders by month.

🛠️ Development Tools

Several tools were used during development to make the process easier and more organized.

- **XAMPP/WAMP:** Local server environments for running PHP and MySQL during development.
- **phpMyAdmin:** Web-based tool for managing the MySQL database.
- **VS Code / Sublime Text:** Code editors used to write and debug HTML, CSS, JavaScript, and PHP.
- **Git (Optional):** Version control system for tracking changes and collaborating.
- **Browser Developer Tools:** Used for debugging frontend issues, checking layouts, and testing responsiveness.

🔒 Security Tools

Security is an important part of the stack.

- **Password Hashing (PHP functions like `password_hash`):** Protects user passwords.
- **Input Validation:** Ensures that forms only accept valid data.
- **SSL Certificates (Planned for Deployment):** Encrypts communication between users and the server.

📊 Testing Tools

Testing ensures that the system works correctly before deployment.

- **Manual Testing:** Each feature (login, cart, customization, admin dashboard) was tested step by step.

- **Browser Testing:** The system was tested on Chrome, Firefox, and Edge to ensure compatibility.
- **Responsive Testing:** Checked layouts on desktop and mobile devices.

Summary

The Tab3a W Hdiyi platform uses a **classic LAMP-style stack** (Linux, Apache, MySQL, PHP) combined with frontend technologies (HTML, CSS, JavaScript). Supporting tools like XAMPP, phpMyAdmin, and VS Code made development easier, while security and testing tools ensured reliability. This stack provides a solid foundation for future improvements, including mobile app development and cloud hosting.

Conclusion

The Tab3a W Hdiyi project represents a complete journey from concept to implementation, showcasing how a modern e-commerce platform can be designed, built, and documented with clarity and precision. At its core, the system provides customers with the ability to browse products, customize items to their liking, and place orders seamlessly, while giving administrators powerful tools to manage inventory, users, orders, and profits. This dual focus — customer experience and administrative control — demonstrates the balance between usability and functionality that every successful platform must achieve.

Throughout the documentation, we explored the **system architecture**, which is built on a classic three-layer model: the client layer (HTML, CSS, JavaScript), the server layer (PHP), and the database layer (MySQL). This architecture ensures that the system is modular, maintainable, and scalable. The **technology stack and tools used** — from XAMPP and phpMyAdmin for development to browser testing and security measures — highlight the practical choices made to ensure smooth development and reliable performance. Each tool was selected not only for its technical capabilities but also for its accessibility to developers and its ability to support future growth.

The **Admin Dashboard** was presented in detail, with its four main tabs — Products, Orders, Users, and Profits — each serving a critical role in system management. The Products tab allows for adding and editing items, the Orders tab tracks customer purchases and delivery statuses, the Users tab manages accounts and roles, and the Profits tab provides financial insights. Together, these features create a comprehensive control center that empowers administrators to keep the system organized and profitable. The storyboard breakdown of each button and input field ensures that both technical and non-technical readers can understand how the system works.

Beyond functionality, the project also emphasizes **Accessibility Features**, ensuring that the platform is usable by all individuals regardless of ability. Clear labels, keyboard navigation, color contrast, alt text for images, and screen reader compatibility make the system inclusive. Accessibility is not treated as an afterthought but as an integral part of the design, reinforcing the professionalism and social responsibility of the project.

The **Deployment & Hosting Plan** outlines how the system can move from a local environment to a live production server. By considering shared hosting, VPS, and cloud hosting options, the plan demonstrates foresight and scalability. Security measures such as SSL certificates, backups, and monitoring ensure that the system will remain reliable and safe once deployed. This section shows that the project is not limited to academic demonstration but is ready for real-world application.

Looking forward, the **Future Improvements** section highlights ambitious yet realistic goals. The development of a mobile application will make the platform more accessible to customers, while payment gateway integration will modernize transactions. Multi-language support, advanced analytics, improved customization tools, and social media integration will expand the system's reach and functionality. Scalability through cloud hosting, community features like reviews and loyalty points, and integration with delivery services will transform Tab3a W Hdiyi into a full-featured e-commerce ecosystem. These improvements show that the project is not static but evolving, with a clear vision for growth.

In conclusion, Tab3a W Hdiyi is more than just a student project — it is a **foundation for a real business platform**. It combines technical rigor with user-centered design, accessibility with scalability, and present functionality with future vision. The documentation captures not only what the system does today but also what it can become tomorrow. By presenting detailed storyboards, architecture, accessibility, deployment, technology stack, and future improvements, this project demonstrates a holistic understanding of software development. It reflects the persistence, organization, and ambition behind its creation, and it stands as a strong example of how thoughtful design and careful planning can lead to a system that is both practical and inspiring.